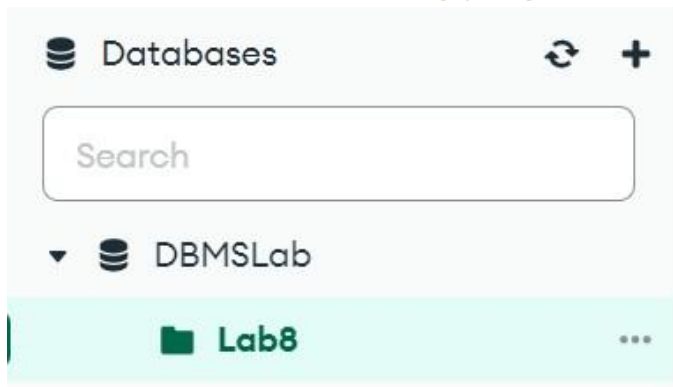


[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: b. Create Collection db.create

[Exp 9: P56]

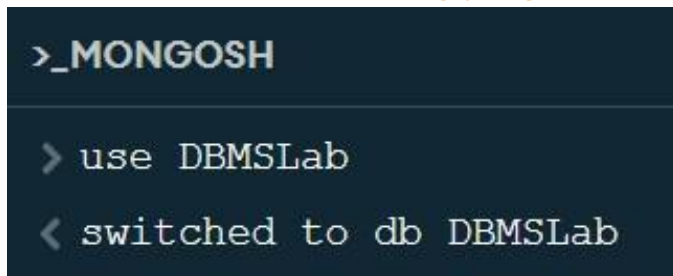


Manual P56 Output (382x219)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: b. Create Collection db.create

[Exp 9: P56]

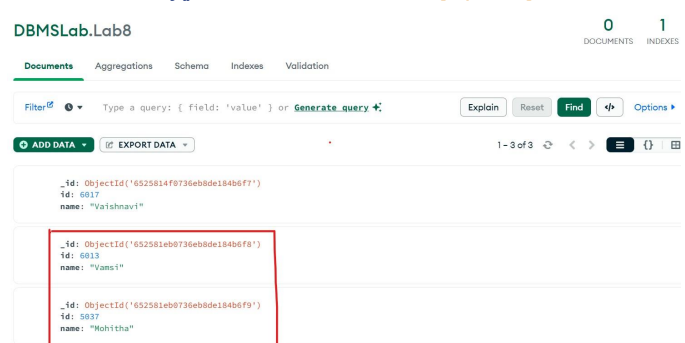


Manual P56 Output (322x129)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: InsertMany() : to insert multi

[Exp 9: P57]



Manual P57 Output (1483x778)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: d. Find and FindOne -

[Exp 9: P58]

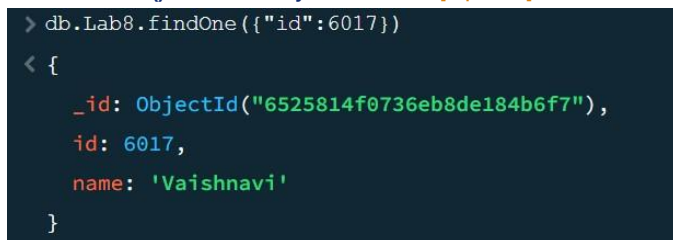


Manual P58 Output (873x282)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: findOne() - To select only one

[Exp 9: P59]

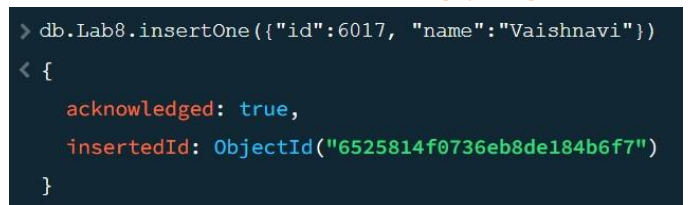


Manual P59 Output (616x217)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: b. Create Collection db.create

[Exp 9: P56]



Manual P56 Output (637x189)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: b. Create Collection db.create

[Exp 9: P56]

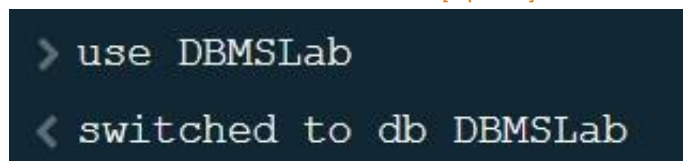


Manual P56 Output (565x133)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: b. Create Collection db.create

[Exp 9: P56]



Manual P56 Output (315x72)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: InsertMany() : to insert multi

[Exp 9: P57]

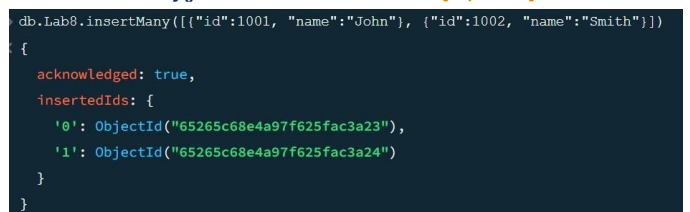


Manual P57 Output (975x288)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: InsertMany() : to insert multi

[Exp 9: P57]

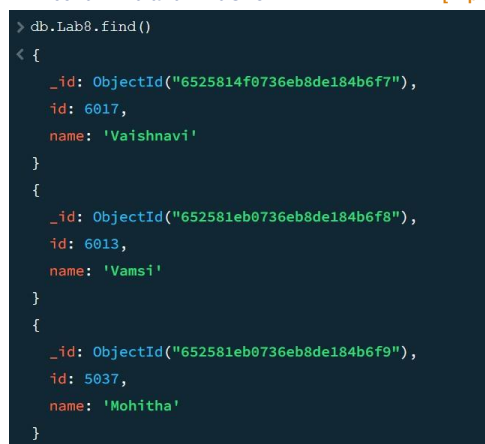


Manual P57 Output (940x286)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: d. Find and FindOne -

[Exp 9: P58]



Manual P58 Output (648x588)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: findOne() - To select only one

[Exp 9: P59]

```
> db.Lab8.find({}, {"name":1})
< {
  _id: ObjectId("6525814f0736eb8de184b6f7"),
  name: 'Vaishnavi'
}
{
  _id: ObjectId("652581eb0736eb8de184b6f8"),
  name: 'Vamsi'
}
{
  _id: ObjectId("652581eb0736eb8de184b6f9"),
  name: 'Mohitha'
}
```

Manual P59 Output (595x478)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: - db.CollectionName.updateOne()

[Exp 9: P60]

```
> db.Lab8.updateMany({}, {$inc:{"id":1}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

Manual P60 Output (541x292)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: g. DeleteOne and DeleteMany -

[Exp 9: P61]

```
> db.Lab8.deleteOne({"id":5039})
< {
  acknowledged: true,
  deletedCount: 1
}
```

Manual P61 Output (430x186)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: db.CollctionName.insertOne()

[Exp 9: P63]

```
> show dbs
< admin      40.00 KiB
  config    108.00 KiB
  local     40.00 KiB
```

Manual P63 Output (259x154)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: - db.CollectionName.updateOne()

[Exp 9: P60]

```
> db.Lab8.updateOne({"id":6017}, {$set:{"name":"Satya Vaishnavi"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P60 Output (835x292)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: - db.CollectionName.updateOne()

[Exp 9: P60]

```
  _id: ObjectId('6525814f0736eb8de184b6f7')
  id: 6017
  name: "Satya Vaishnavi"
```

Manual P60 Output (559x127)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: g. DeleteOne and DeleteMany -

[Exp 9: P61]

```
  _id: ObjectId('6525814f0736eb8de184b6f7')
  id: 6019
  name: "Satya Vaishnavi"

  _id: ObjectId('652581eb0736eb8de184b6f8')
  id: 6015
  name: "Vamsi"

  _id: ObjectId('652581eb0736eb8de184b6f9')
  id: 5039
  name: "Mohitha"

  _id: ObjectId('65265c68e4a97f625fac3a23')
  id: 1003
  name: "John"
```

Manual P61 Output (583x544)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: deleteMany() - The deleteMany()

[Exp 9: P62]

DBMSLab.Lab8

Documents Aggregations Schema Indexes Validation

Filter Type a query: { field: 'value' } or Generate query

ADD DATA EXPORT DATA

```
  _id: ObjectId("6525814f0736eb8de184b6f7")
  id: 6019
  name: "Satya Vaishnavi"

  _id: ObjectId("652581eb0736eb8de184b6f8")
  id: 6015
  name: "Vamsi"

  _id: ObjectId("65265c68e4a97f625fac3a23")
  id: 1003
  name: "John"

  _id: ObjectId("65265c68e4a97f625fac3a24")
  id: 1004
  name: "Smith"
```

Manual P62 Output (982x906)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: db.CollctionName.insertOne()

[Exp 9: P63]

```
> db.Lab8.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 4
}
```

Manual P63 Output (343x183)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: db.CollcetionName.insertOne({"

[Exp 9: P63]

ADD DATA EXPORT DATA

0 - 0 of 0



This collection has no data

It only takes a few seconds to import data from a JSON or CSV file.

Import Data

Manual P63 Output (1503x610)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[Exp 10: P64]

```
db.Lab9.insertMany([{"id":1001, "name":"Arjun", "marks":70}, {"id":1002, "name":"Keerthi", "marks":80}, {"id": 1002, "name":"Rudra", "marks":75}, {"id":1004, "name":"Ridhima", "marks":60}, {"id":1005, "name":"Sanjana", "marks":50}, {"id":1006, "name":"Vaishnavi", "marks":95}, {"id":1007, "name":"Vamsi", "marks":90}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("6526c663d0b2b55b14aca5f1"),
    '1': ObjectId("6526c663d0b2b55b14aca5f2"),
    '2': ObjectId("6526c663d0b2b55b14aca5f3"),
    '3': ObjectId("6526c663d0b2b55b14aca5f4"),
    '4': ObjectId("6526c663d0b2b55b14aca5f5"),
    '5': ObjectId("6526c663d0b2b55b14aca5f6"),
    '6': ObjectId("6526c663d0b2b55b14aca5f7")
  }
}
```

Manual P64 Operator (1552x544)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[Exp 10: P64]

```
_id: ObjectId('6526c663d0b2b55b14aca5f4')
id: 1004
name: "Ridhima"
marks: 60
```

```
_id: ObjectId('6526c663d0b2b55b14aca5f5')
id: 1005
name: "Sanjana"
marks: 50
```

```
_id: ObjectId('6526c663d0b2b55b14aca5f6')
id: 1006
name: "Vaishnavi"
marks: 95
```

Manual P64 Operator (558x483)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({"id": {\$ne: 10

[Exp 10: P65]

```
> db.Lab9.find({"id":{$eq: 1006}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual P65 Operator (565x258)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: \$gt: Value is greater than ano

[Exp 10: P66]

```
> db.Lab9.find({"id": {$ne: 1006}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
```

Manual P66 Operator (595x255)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[Exp 10: P64]

```
> use DBMSLab
< switched to db DBMSLab
```

Manual P64 Operator (337x73)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[Exp 10: P64]

```
> db.createCollection("Lab9")
< { ok: 1 }
```

Manual P64 Operator (364x77)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[Exp 10: P64]

```
_id: ObjectId('6526c663d0b2b55b14aca5f1')
id: 1001
name: "Arjun"
marks: 70
```

```
_id: ObjectId('6526c663d0b2b55b14aca5f2')
id: 1002
name: "Keerthi"
marks: 80
```

```
_id: ObjectId('6526c663d0b2b55b14aca5f3')
id: 1003
name: "Rudra"
marks: 75
```

Manual P64 Operator (526x460)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({"id": {\$ne: 10

[Exp 10: P65]

```
_id: ObjectId('6526c663d0b2b55b14aca5f7')
id: 1007
name: "Vamsi"
marks: 90
```

Manual P65 Operator (528x171)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({"id": {\$ne: 10

[Exp 10: P65]

```
> db.Lab9.find({"id": 1006})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual P65 Operator (618x256)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: \$gt: Value is greater than ano

[Exp 10: P66]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual P66 Operator (577x651)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: \$gt: Value is greater than ano

[Exp 10: P66]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual P66 Operator (562x430)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: \$gte: Value is greater than or

[Exp 10: P67]

```
> db.Lab9.find({"marks":{$gte: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual P67 Operator (570x475)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: \$gte: Value is greater than or

[Exp 10: P67]

```
> db.Lab9.find({"marks":{$gte: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual P67 Operator (592x685)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: \$lte: Value is less than or eq

[Exp 10: P68]

```
> db.Lab9.find({"marks":{$lte: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual P68 Operator (559x694)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: \$lte: Value is less than or eq

[Exp 10: P68]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual P68 Operator (562x219)

[CUT] CUT ALONG DASHED LINE [GUT]

EXP 10: Query Operator #16 (P69)

[Exp 10: P69]

```
> db.Lab9.find({"marks":{$lte: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
```

Manual P69 Operator (571x691)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: Query Operator #17 (P69)**[Exp 10: P69]**

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual P69 Operator (580x433)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$in: Value is matched within a**[Exp 10: P70]**

```
> db.Lab9.find({"id":{$in:[ 1001, 1005 ]}}, { _id: 1})

< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1")
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5")
}
```

Manual P70 Operator (639x294)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$and: Returns documents where**[Exp 10: P71]**

```
> db.Lab9.find({"id":{$in:[ 1001, 1005 ]}}, {}, { _id: 1})

< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual P71 Operator (694x511)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$or: Returns documents where e**[Exp 10: P73]**

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual P73 Operator (643x217)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$in: Value is matched within a**[Exp 10: P70]**

```
> db.Lab9.find({"id":{$in:[ 1001, 1005 ]}}, { _id: 0})

< {
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual P70 Operator (642x441)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$and: Returns documents where**[Exp 10: P71]**

```
> db.Lab9.find({$and: [{"id": {$ne: 1001}}, {"marks": {$exists: true}}]})

< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
```

Manual P71 Operator (868x478)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: "id" db.Lab9.find({"id": {\$eq:**[Exp 10: P72]**

```
> db.Lab9.find({"id": {$eq: 1006, $exists: true}})

< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual P72 Operator (628x258)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: "id" db.Lab9.find({"id": {\$eq:**[Exp 10: P72]**

```
> db.Lab9.find({$and: [{"id": {$ne: 1001}}, {"subject": {$exists: false}}]})

< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual P72 Operator (901x697)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: \$or: Returns documents where e [Exp 10: P73]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual P73 Operator (652x441)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: Query Operator #28 (P75) [Exp 10: P75]

```
> db.Lab9.find({$nor: [{"id": 1001}, {"marks": true}]})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual P75 Operator (658x691)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: { \$not: { \$gt: 1003 } } [Exp 10: P77]

```
> db.Lab9.find({"id": {$not: {$gt: 1003 }}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
```

Manual P77 Operator (579x687)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: - db.Lab9.find({\$or: [{"marks" [Exp 10: P74]

```
> db.Lab9.find({"id": {$eq: 1006, $exists: true}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual P74 Operator (628x258)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: - db.Lab9.find({\$or: [{"marks" [Exp 10: P74]

```
> db.Lab9.find({$or: [{"marks": {$lt: 70}}, {"id": {$eq: 1006}]}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual P74 Operator (790x691)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: Query Operator #29 (P75) [Exp 10: P75]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual P75 Operator (612x646)

-- [CUT] CUT ALONG DASHED LINE [GUT] --

EXP 10: { \$not: { \$gt: 1003 } } [Exp 10: P77]

```
> db.Lab9.find({$and: [{"id": {$eq: 1001}}, {"marks": {$eq: 70}]}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
```

Manual P77 Operator (814x258)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: \$regex: Allows the use of regu

[Exp 10: P78]

```
> db.sampledata.find({"name": {$regex: /^E/}})
< {
  _id: '5f1234567890123456789015',
  employeeID: 1004,
  name: 'Emily Davis',
  age: 29,
  isEmployee: true,
  position: 'UI/UX Designer',
  salary: 70000.5,
  joinDate: '2020-06-05T00:00:00Z',
  skills: [
    'Adobe XD',
    'Sketch',
    'User Research'
  ],
  address: {
    street: '101 Pine St',
    city: 'Othercity',
    state: 'TX',
    zip: '98765'
  }
}
```

Manual P78 Operator (637x796)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Query Operator #33 (P79)

[Exp 10: P79]

```
{
  _id: '5f1234567890123456789022',
  employeeID: 1011,
  name: 'Emma Hernandez',
  age: 29,
  isEmployee: true,
  position: 'Product Manager',
  salary: 92000.5,
  joinDate: '2017-02-14T00:00:00Z',
  skills: [
    'Product Roadmapping',
    'Market Research',
    'User Experience'
  ],
  address: {
    street: '888 Elmwood St',
    city: 'Elsewhere',
    state: 'UT',
    zip: '22345'
  }
}
```

Manual P79 Operator (565x759)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Query Operator #34 (P79)

[Exp 10: P79]

```
{
  _id: '5f1234567890123456789031',
  employeeID: 1020,
  name: 'Ella Turner',
  age: 30,
  isEmployee: true,
  position: 'Sales Manager',
  salary: 88000.25,
  joinDate: '2008-01-20T00:00:00Z',
  skills: [
    'Sales Strategy',
    'Client Relationship',
    'Negotiation'
  ],
  address: {
    street: '999 Elm St',
    city: 'Nowhere',
    state: 'MA',
    zip: '99887'
  }
}
```

Manual P79 Operator (505x763)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Query Operator #35 (P81)

[Exp 10: P81]

```
> db.sampledata.createIndex({position: "text"})
< position_text
> db.sampledata.find({$text:{$search: "Developer"}})
< {
  _id: '5f1234567890123456789013',
  employeeID: 1002,
  name: 'Jane Smith',
  age: 28,
  isEmployee: true,
  position: 'Web Developer',
  salary: 65000.75,
  joinDate: '2021-03-20T00:00:00Z',
  skills: [
    'HTML',
    'CSS',
    'JavaScript'
  ],
  address: {
    street: '456 Elm St',
    city: 'Sometown',
    state: 'NV',
    zip: '67890'
  }
}
```

Manual P81 Operator (652x838)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Query Operator #36 (P81)

[Exp 10: P81]

```
}
{
  _id: '5f1234567890123456789019',
  employeeID: 1008,
  name: 'Ava Hall',
  age: 27,
  isEmployee: true,
  position: 'Mobile App Developer',
  salary: 72000.75,
  joinDate: '2021-01-02T00:00:00Z',
  skills: [
    'React Native',
    'Swift',
    'Android'
  ],
  address: {
    street: '555 Pineapple St',
    city: 'Somewhere',
    state: 'AZ',
    zip: '11234'
  }
}
```

Manual P81 Operator (571x796)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$inc: Increments the field val

[Exp 11: P83]

```
> db.sampledata.updateOne({"name": "Emily Davis"}, {$currentDate: {"joinDate":true}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P83 Update (1029x295)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - It will rename the field of

[Exp 11: P84]

```
> db.sampledata.updateOne({"name": "Emily Davis"}, {$currentDate: {"joinDate":true}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.sampledata.updateOne({"name":"Emily Davis"}, {$inc:{"age": -1, "salary": 5000}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P84 Update (1026x579)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$inc: Increments the field val

[Exp 11: P83]

```
  _id: "5f1234567890123456789015"
 employeeID: 1004
  name: "Emily Davis"
  age: 29
 isEmployee: true
 position: "UI/UX Designer"
 salary: 70000.5
 joinDate: 2023-11-03T11:04:38.482+00:00
▶ skills: Array (3)
▶ address: Object
```

Manual P83 Update (730x316)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - It will rename the field of

[Exp 11: P84]

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emily Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 75000.5
joinDate: 2023-11-03T11:04:38.482+00:00
skills: Array (3)
address: Object
```

Manual P84 Update (604x292)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$set: Sets the value of a fiel

[Exp 11: P85]

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emily Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 75000.5
skills: Array (3)
address: Object
JoiningDate: 2023-11-03T11:04:38.482+00:00
```

Manual P85 Update (577x319)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: 'new value only' like shown in

[Exp 11: P87]

```
> db.sampledata.updateOne({"name":"Sophia Brown"}, {$set: {"skills":["Python","R","Machine Learning","C+"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P87 Update (1333x301)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$unset: Removes the field from

[Exp 11: P88]

```
> db.sampledata.updateOne({"name":"Sophia Brown"}, {$set: {"skills":["C+"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P88 Update (994x298)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$unset: Removes the field from

[Exp 11: P88]

```
_id: "5f1234567890123456789017"
employeeID: 1006
name: "Sophia Brown"
age: 31
isEmployee: true
position: "Data Scientist"
salary: 90000
joinDate: "2017-07-20T00:00:00Z"
skills: Array (1)
  0: "C++"
address: Object
  street: "333 Cedar St"
  city: "Othertown"
  state: "IL"
  zip: "33445"
```

Manual P88 Update (529x433)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$set: Sets the value of a fiel

[Exp 11: P85]

```
> db.sampledata.updateOne({"name":"Emily Davis"}, {$set: {"name":"Emilia Davis", "salary":10000}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P85 Update (1198x292)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$set: Sets the value of a fiel

[Exp 11: P85]

```
> db.sampledata.updateOne({"name":"Emily Davis"}, {$rename: {"joinDate":"JoiningDate"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P85 Update (1039x289)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - db.sampledata.updateOne({"na

[Exp 11: P86]

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emilia Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 10000
skills: Array (3)
address: Object
JoiningDate: 2023-11-03T11:04:38.482+00:00
```

Manual P86 Update (627x312)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: 'new value only' like shown in

[Exp 11: P87]

```
_id: "5f1234567890123456789017"
employeeID: 1006
name: "Sophia Brown"
age: 31
isEmployee: true
position: "Data Scientist"
salary: 90000
joinDate: "2017-07-20T00:00:00Z"
skills: Array (4)
  0: "Python"
  1: "R"
  2: "Machine Learning"
  3: "C++"
address: Object
  street: "333 Cedar St"
  city: "Othertown"
  state: "IL"
  zip: "33445"
```

Manual P87 Update (478x520)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$addToSet: Adds distinct eleme

[Exp 11: P89]

```
> db.sampledata.updateOne({"name":"Emilia Davis"}, {$unset: {"JoiningDate":" ", "address":" "}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P89 Update (1147x297)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: \$addToSet: Adds distinct eleme [Exp 11: P89]

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emilia Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 10000
▼ skills: Array (3)
  0: "Adobe XD"
  1: "Sketch"
  2: "User Research"
```

Manual P89 Update (490x330)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: \$pull: Removes all elements fr [Exp 11: P91]

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$pull: {"skills": -1}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P91 Update (928x300)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: - {\$pull: {<field1>: <value>|co [Exp 11: P92]

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$pull: {"skills": {$in: ["Swift", "Android", "Java"]}}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P92 Update (1261x300)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: - {\$pull: {<field1>: <value>|co [Exp 11: P92]

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
▼ skills: Array (empty)
▼ address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual P92 Update (454x400)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 12: MongoDB \$match and \$group Pipe [Exp 12: Left Page (A)]

```
> db.sales.aggregate([ { $match: { status: 'Co
[
  { "_id": "Electronics", "totalRevenue": 4500
  { "_id": "Furniture", "totalRevenue": 1850
  { "_id": "Stationery", "totalRevenue": 3200
]
```

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 12: MongoDB \$lookup Stage - Relati [Exp 12: Left Page (A)]

```
> db.orders.aggregate([ { $lookup: { from: 'cu
[
  { "_id": ObjectId("60a1..."), "order_no": "0
  { "_id": ObjectId("60a2..."), "order_no": "0
]
```

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: \$pop: Removes the first or las [Exp 11: P90]

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$addToSet: {"skills": "Java"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P90 Update (994x295)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: \$pop: Removes the first or las [Exp 11: P90]

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
▼ skills: Array (4)
  0: "React Native"
  1: "Swift"
  2: "Android"
  3: "Java"
▼ address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual P90 Update (480x489)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: \$pull: Removes all elements fr [Exp 11: P91]

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
▼ skills: Array (3)
  0: "Swift"
  1: "Android"
  2: "Java"
▼ address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual P91 Update (523x484)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: Update Operation #21 (P93) [Exp 11: P93]

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$push: {"skills": ["React Native", "Swift", "Android", "JavaScript", "C++"]}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual P93 Update (1518x285)

- [CUT] CUT ALONG DASHED LINE [CUT] -

EXP 11: Update Operation #22 (P93) [Exp 11: P93]

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
▼ skills: Array (1)
  0: Array (5)
    0: "React Native"
    1: "Swift"
    2: "Android"
    3: "JavaScript"
    4: "C++"
▼ address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual P93 Update (459x532)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 12: MongoDB \$project & \$multiply C

[Exp 12: Left Page (P)]

```
> db.sales.aggregate([ { $project: { item: 1,
[
  { "_id": ObjectId("672c4..."), "item": "appl
  { "_id": ObjectId("672c5..."), "item": "bana
  { "_id": ObjectId("672c6..."), "item": "oran
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 13: Creating Compound & Full-Text

[Exp 13: Left Page (I)]

```
> db.articles.createIndex({ content: 'text', t
{
  "numIndexesBefore": 1,
  "numIndexesAfter": 2,
  "createdCollectionAutomatically": false,
  "ok": 1
}
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 13: MongoDB Wildcard Regex & Range

[Exp 13: Left Page (W)]

```
> db.employees.find({ first_name: { $regex: '^
[
  { "first_name": "John", "last_name": "Doe",
  { "first_name": "Joan", "last_name": "Smith"
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 12: MongoDB \$sort, \$limit & \$count

[Exp 12: Left Page (P)]

```
> db.sales.aggregate([ { $sort: { price: -1 }
{ "topExpensiveItems": 3 }
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 12: MongoDB \$unwind & \$addFields S

[Exp 12: Left Page (U)]

```
> db.inventory.aggregate([ { $unwind: '$tags'
[
  { "_id": 1, "item": "ABC", "tags": "app", "f
  { "_id": 1, "item": "ABC", "tags": "web", "f
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 13: MongoDB Full-Text Search Execu

[Exp 13: Left Page (T)]

```
> db.articles.find({ $text: { $search: 'databa
[
  { "_id": ObjectId("61a..."), "title": "Intro
  { "_id": ObjectId("61b..."), "title": "Moder
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 13: Dropping Indexes & Index Inspe

[Exp 13: Left Page (I)]

```
> db.employees.getIndexes(); db.employees.drop
{ "nIndexesWas": 3, "ok": 1 }
```